

PROJECT BACKGROUND AND CONCEPT

BACKGROUND FROM RESEARCH THESIS

What is the project about? Transforming Thika's metal works Juakali spaces into safer, more productive engines of local growth. This is through first understanding the relationship between spatial outcomes and material logic. Specifically in Kigandaini, the main metal works juakali zone. Thereafter, applying the suitable logic in material selection to achieve the desired spatial outcomes

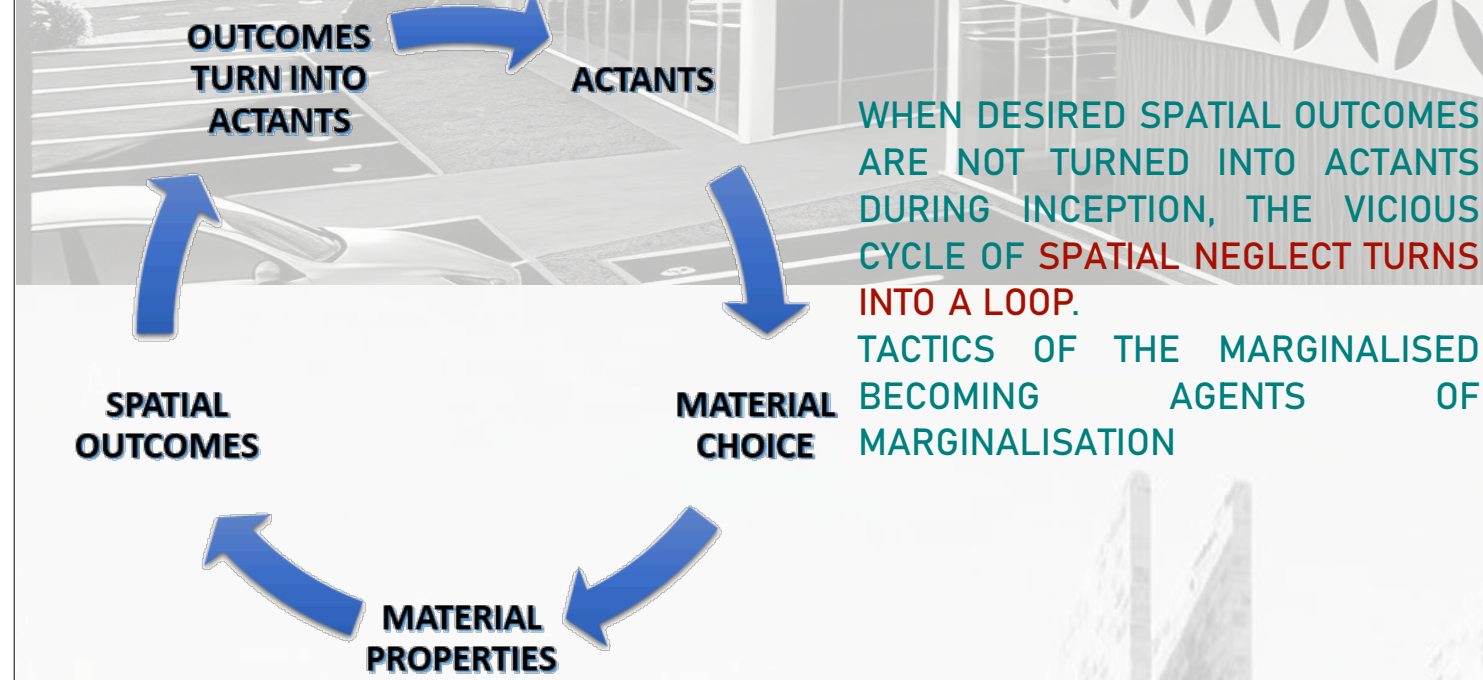
The problem? The existing metal works spaces face various issues like:

- Poor ventilation, lighting and thermal comfort.
- Difficulty in expansion leading to obstruction and encroachment on pedestrian and vehicular paths
- Safety risks like cuts and injuries sustained from the existing materials
- Security risk especially for raw materials and finished goods.
- Poor structural stability
- Water ingress into space
- Poor aesthetic quality

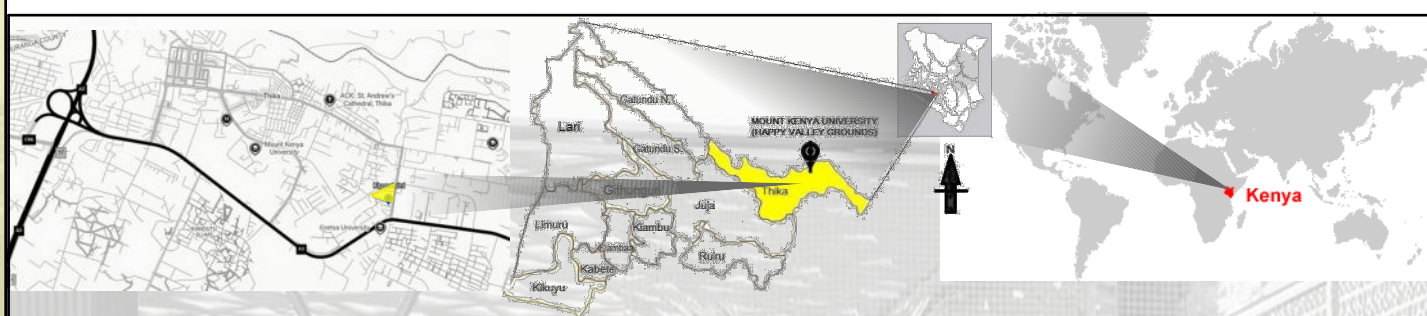


Images showing the various problems in Kigandaini metal works workshops. source- Maria Gicheha 2025

Why solve it? The Juakali metalwork sector in Kigandaini, Thika and Kenya at large plays a vital role in the local economy through employment, affordable production, and creative problem-solving, yet its potential is limited by unstructured spaces shaped by the use of low-cost and often substandard materials.



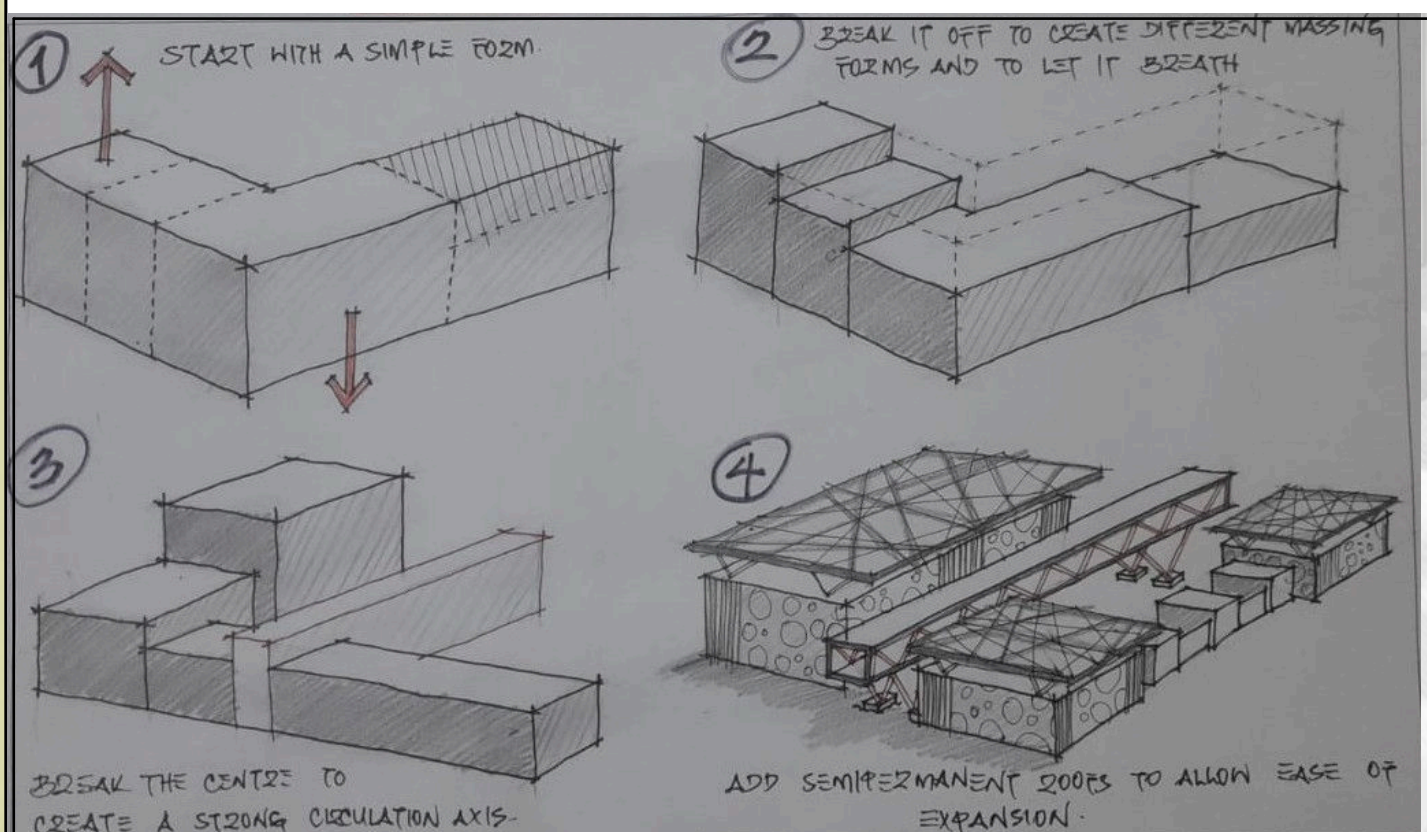
SITE LOCATION



ZONING AND SPACE PLANNING



FORM DEVELOPMENT

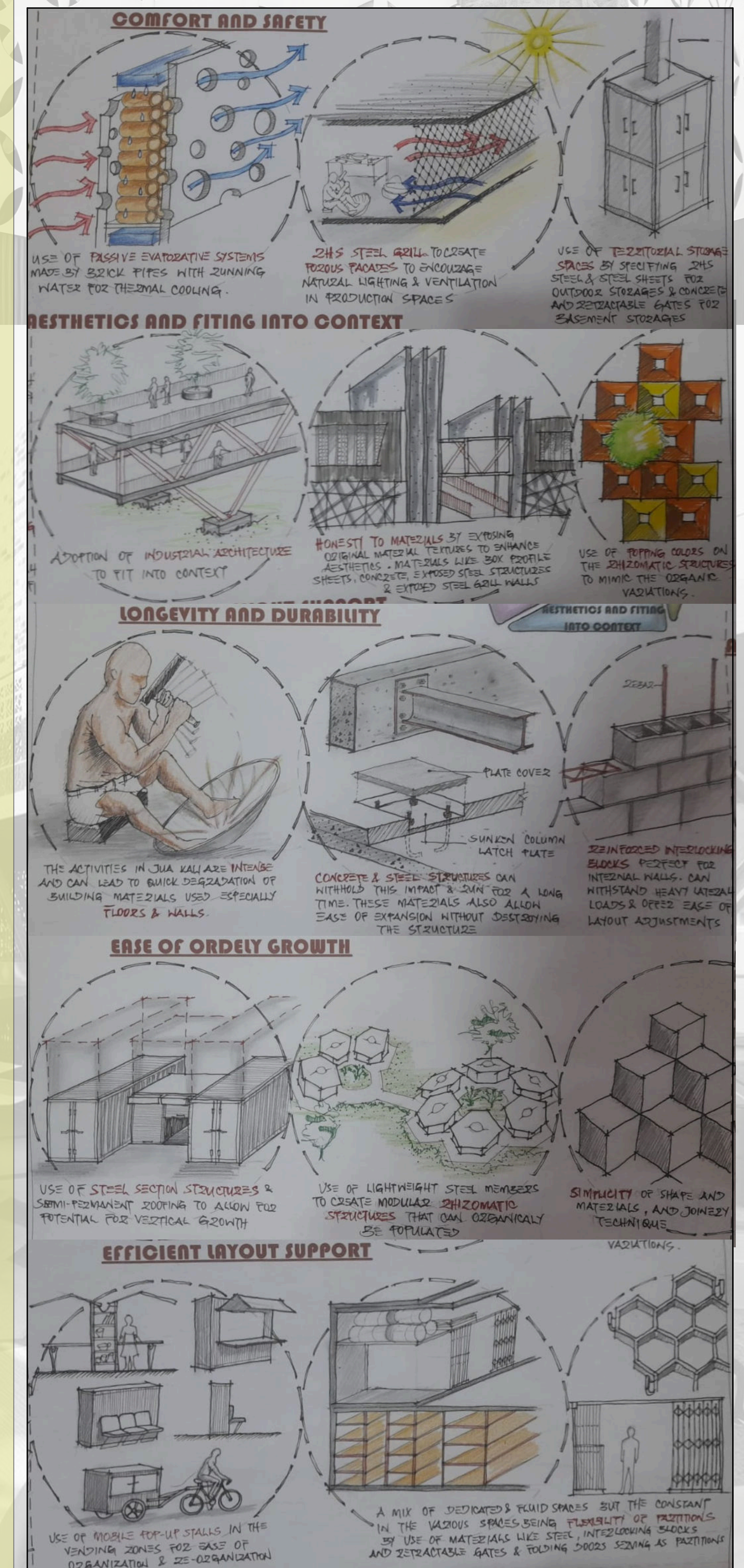
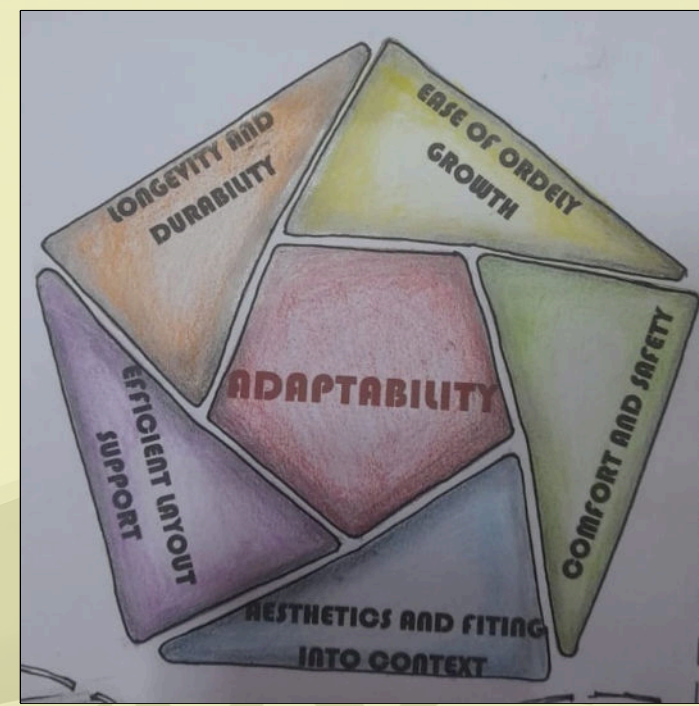


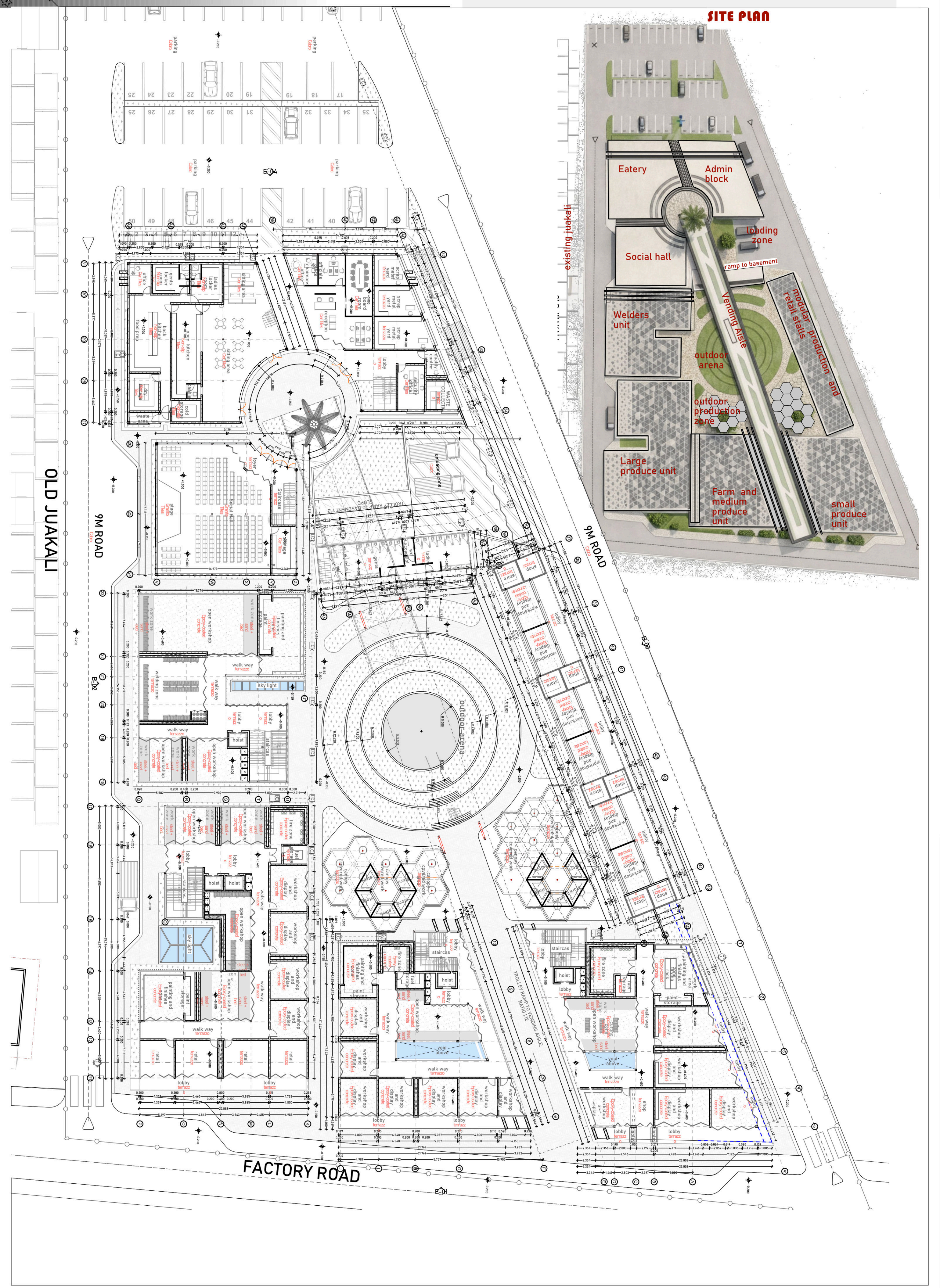
CONCEPT: ADAPTABILITY

CONCEPT PHILOSOPHY: ADAPTABILITY. JUAKALIS AS A BECOMING THROUGH FUSION OF TACTICS AND STRATEGIES

CONCEPT STATEMENT: BY FRAMING ADAPTABILITY AS BECOMING INFRASTRUCTURE, I ARGUE THAT JUAKALI ARCHITECTURE SHOULD NOT ONLY BE ECONOMIC BUT ALSO CONSTRUCTED WITH **CHANGE, LONGEVITY, SAFETY AND COMFORT** AS PRIMARY OPERATING LOGICS

- WITH THE METHOD BEING HEAVY ON MATERIAL TECHNOLOGY GUIDED BY THE DESIRED OUTCOMES AS ACTANTS BEHIND LOGIC OF CHOICE, JUAKALI ARCHITECTURE CAN BE NON-CHAOTIC SYSTEMS THAT CAN BE ARTICULATED AND REARTICULATED, POPULATED AND DISMANTLED, WITHOUT COMPROMISING THE INTENDED SPATIAL OUTCOMES.





SITE PLAN

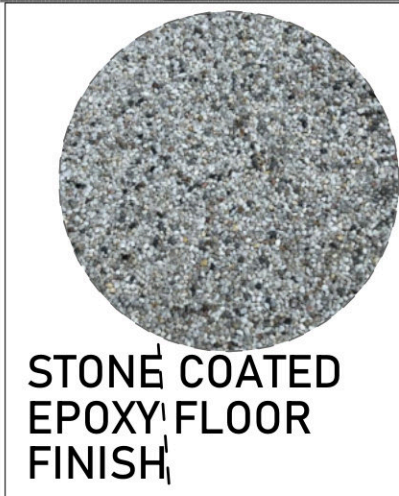
existing juakali

9M ROAD

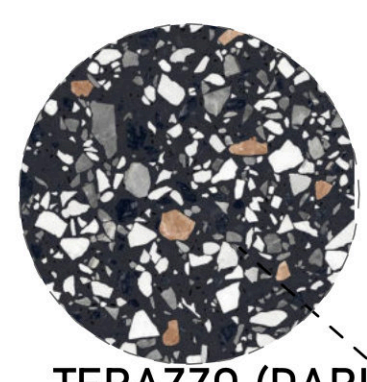
FACTORY ROAD

OLD JUAKALI

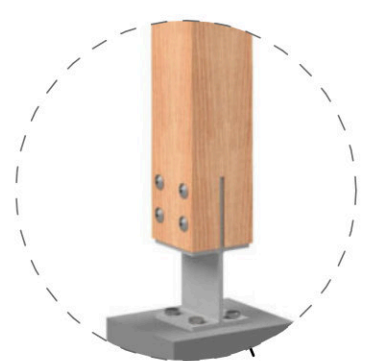
9M ROAD



STONE COATED
EPOXY FLOOR
FINISH



TERAZZO (DARK
COLOURED)



VERTICAL ENGINEERED
TIMBER FINIS



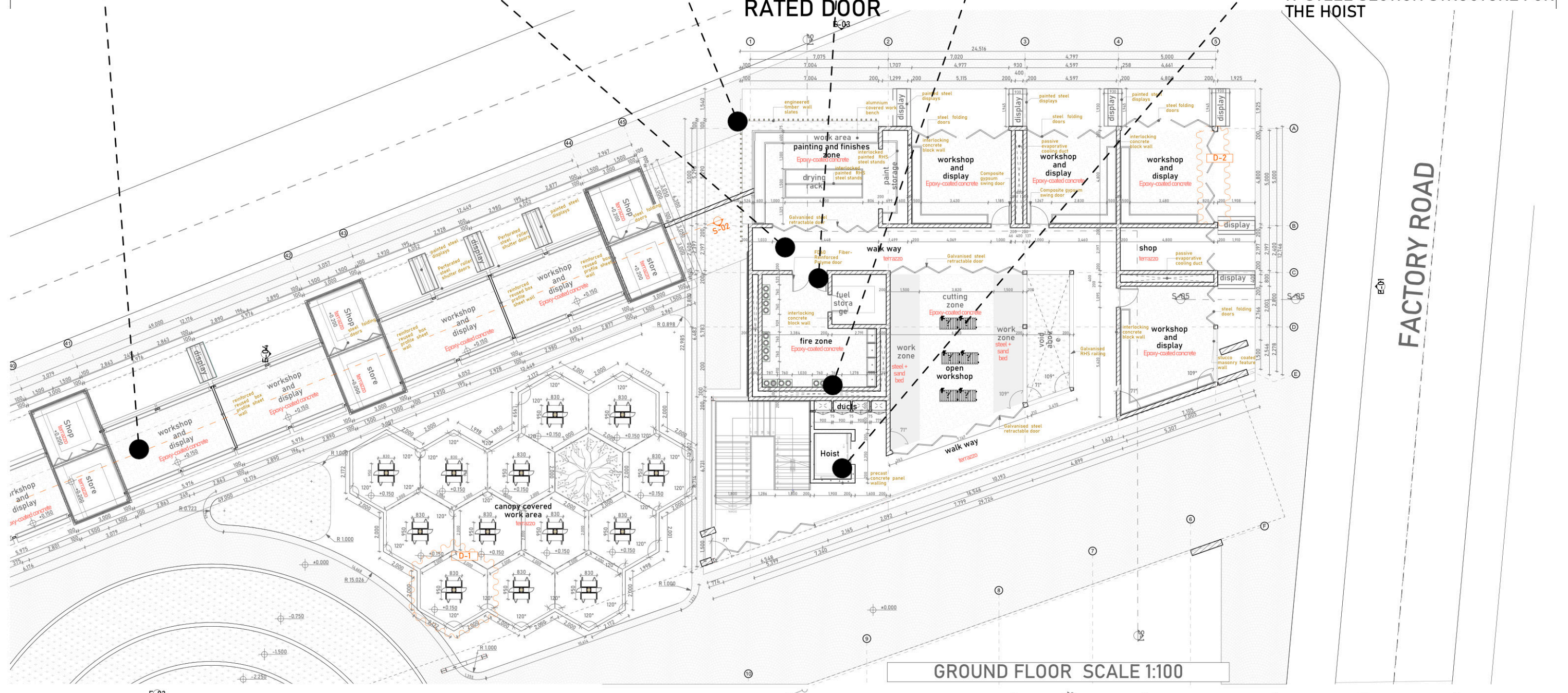
REIGNFORCED FIBER
POLYMER CLASS A2 FIRE
RATED DOOR



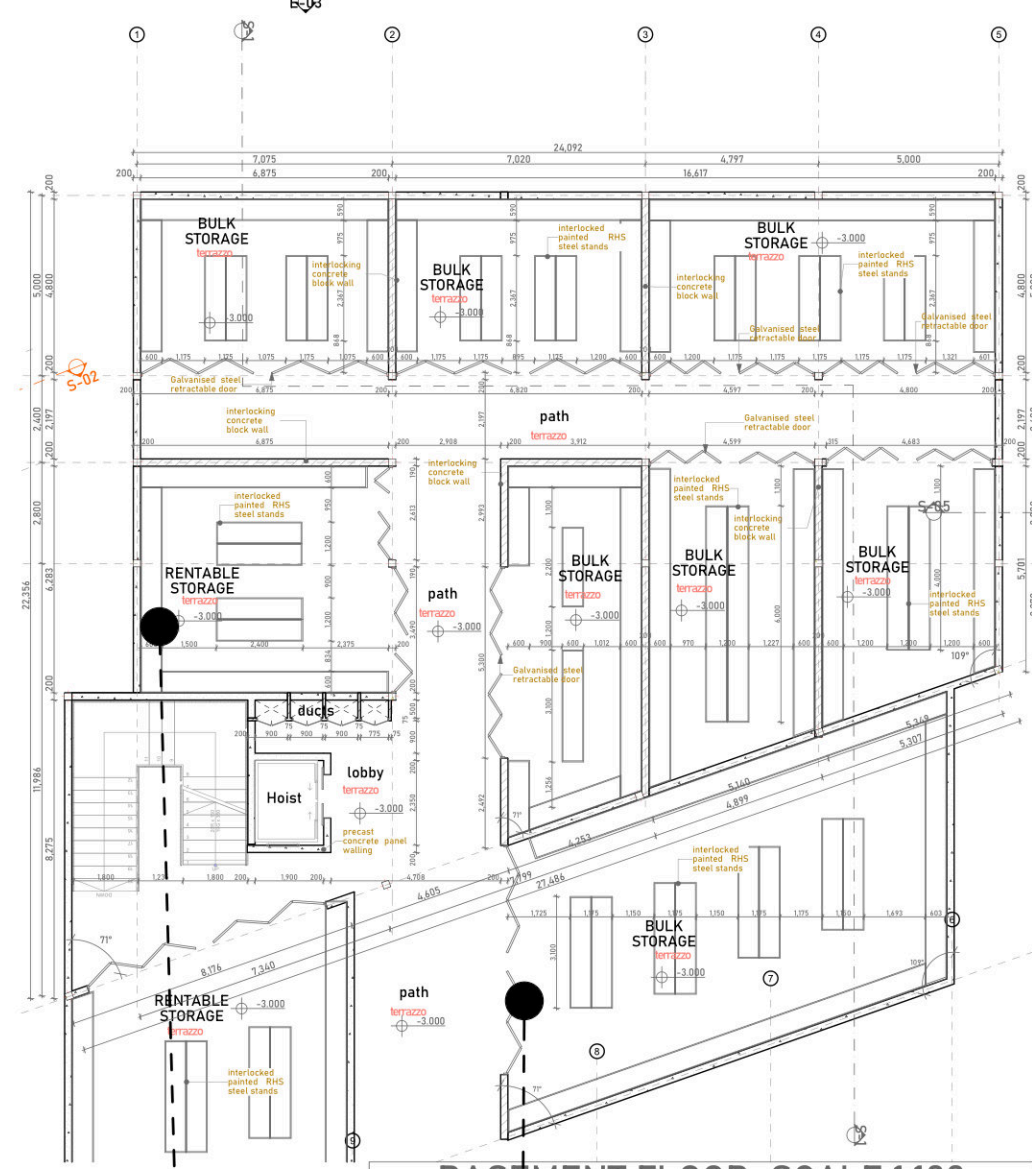
STEEL FIRING
CHAMBER AND
CHIMNEY



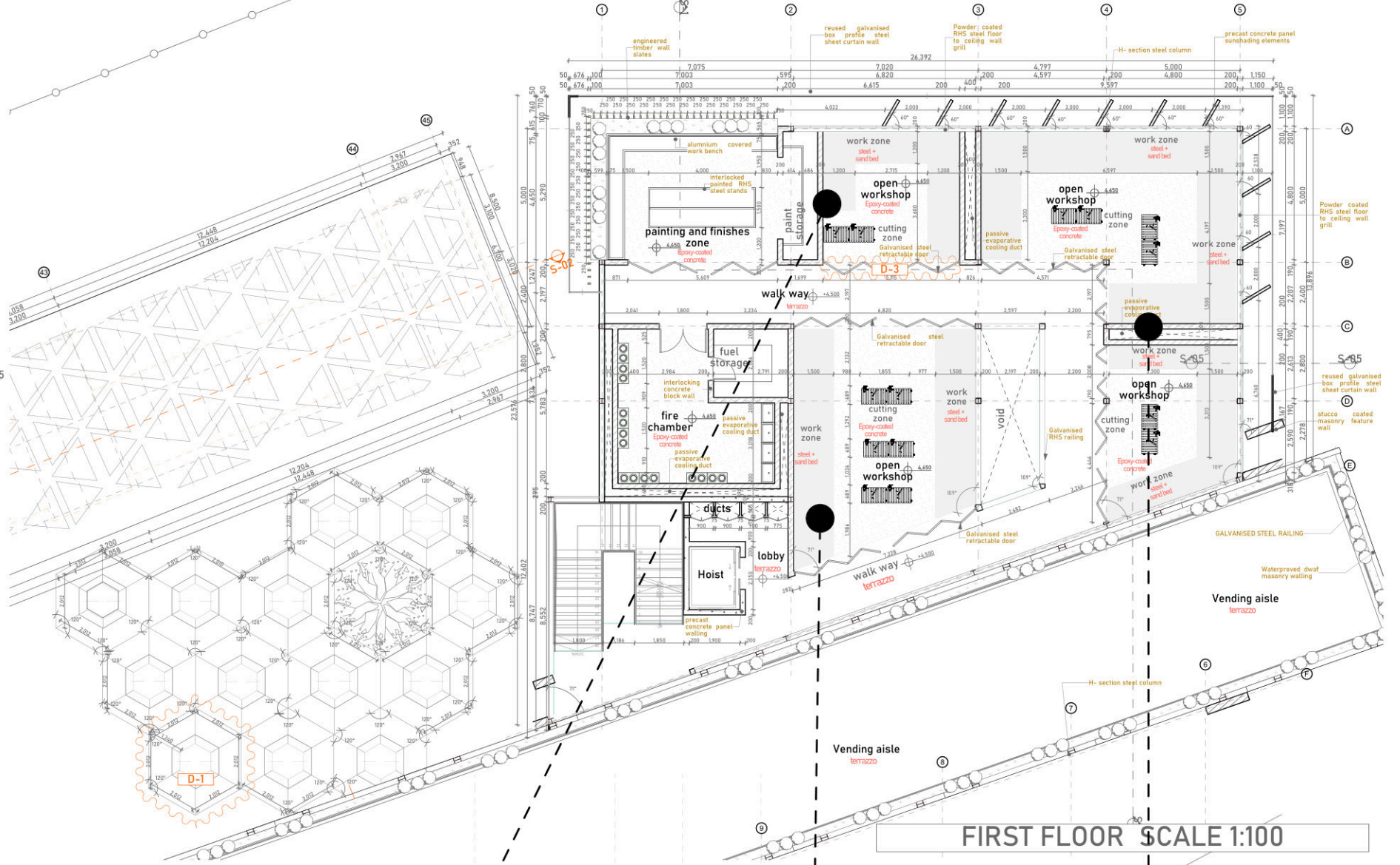
H-STEEL SECTION STRUCTURE FOR
THE HOIST



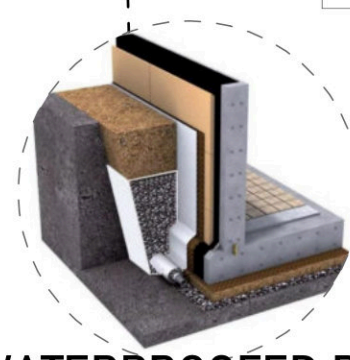
GROUND FLOOR SCALE 1:100



BASEMENT FLOOR SCALE 1:100



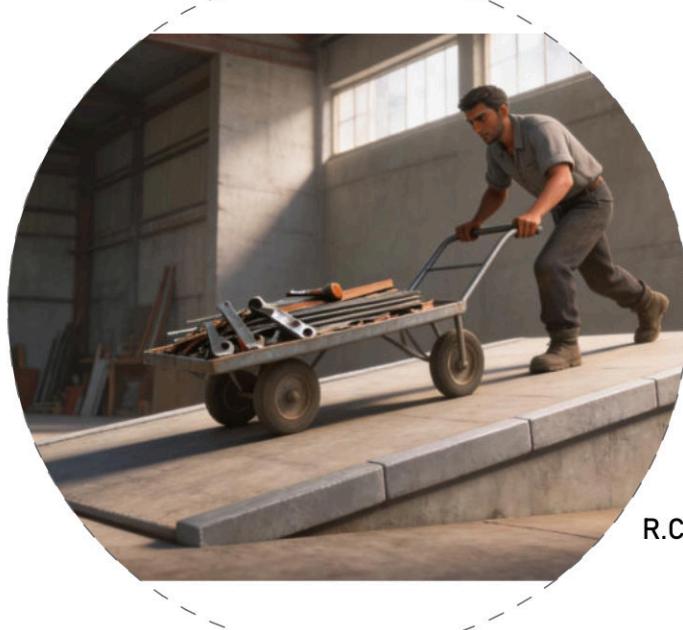
FIRST FLOOR SCALE 1:100



WATERPROOFED R.C
RETAINING WALL



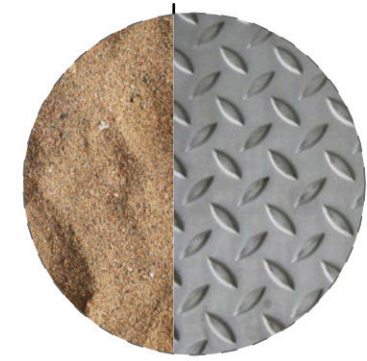
STEEL
COLLAPSIBLE
DOOR



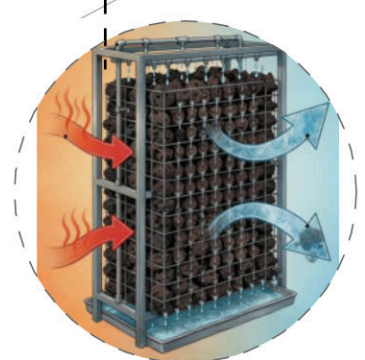
R.C TROLLEY RAMP



REIGNFORCED
INTERLOCKING CONCRETE
BLOCKS



CHECKERED STEEL AND
SAND BED FLOOR
COMPOSITE



STAINLESS STEEL
MESH AND CHARCOAL
COMPOSITE COOLING

water proofing aluminum capping on Ndarugu stone masonry wall with english bond key finish defining primary entry node.

Perforated mild steel roofing over translucent polycarbonate sheets fixed to steel truss system; assembly allows controlled daylighting and future vertical extension.

CHS columns on base plates anchored to steel deck floor supporting semi-permanent roof.

Powder-coated steel railings bolted to steel deck floor for edge protection.

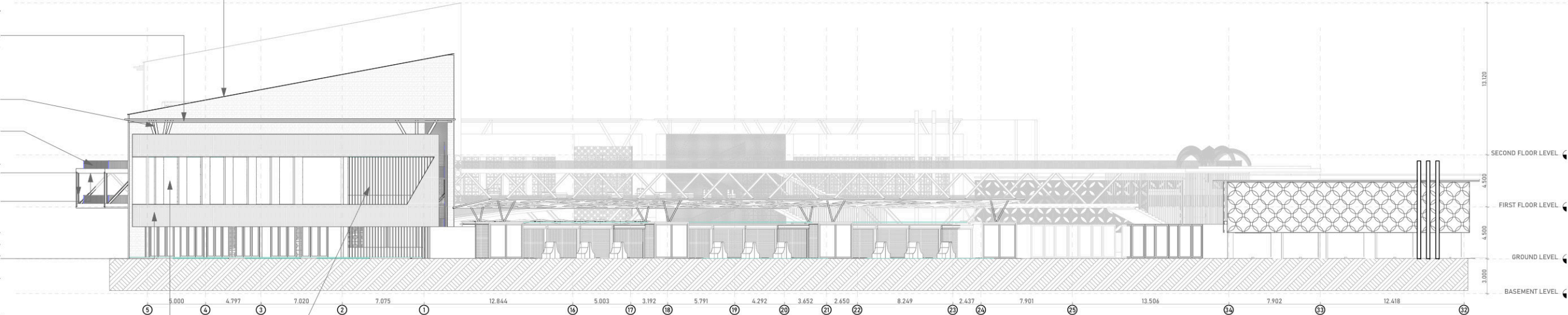
Exposed red-painted H-section steel members supporting elevated vending aisle bridge.

Terrazzo floor finish on reinforced concrete topping cast over steel deck

Reused galvanised box profile steel sheets bolted to RHS structural members for demountable industrial wall enclosure; allows controlled daylighting and future vertical extension.

Precast concrete vertical fins bolted to RHS steel members to reduce solar gain while maintaining ventilation.

Engineered timber wall slats bolted to H-section steel beams forming a ventilated screen to the painting area.



Perforated mild steel roofing over translucent polycarbonate sheets fixed to steel truss system; assembly allows controlled daylighting and future vertical extension.

CHS columns on base plates anchored to steel deck floor supporting semi-permanent roof.

water proofing aluminum capping on Ndarugu stone masonry wall with english bond key finish defining primary entry node.

Powder-coated CHS steel tube floor-to-ceiling grill providing passive ventilation, daylight penetration, and visual security.

Precast concrete vertical fins bolted to RHS steel members to reduce solar gain while maintaining ventilation.

Powder-coated steel railings bolted to steel deck floor for edge protection.

Terrazzo floor finish on reinforced concrete topping cast over steel deck

Exposed red-painted H-section steel members supporting elevated vending aisle bridge.

Reused galvanised box profile steel sheets bolted to RHS structural members for demountable industrial wall enclosure; allows controlled daylighting and future vertical extension.

Engineered timber wall slats bolted to H-section steel beams forming a ventilated screen to the painting area.

ELEVATION E2 SCALE- 1:200



water proofing aluminum capping on Ndarugu stone masonry wall with english bond key finish defining primary entry node.

Exposed red-painted H-section steel members supporting elevated vending aisle bridge.

CHS columns on base plates anchored to steel deck floor supporting semi-permanent roof.

Perforated mild steel roofing over translucent polycarbonate sheets fixed to steel truss system; assembly allows controlled daylighting and future vertical extension.

Reused galvanised box profile steel sheets bolted to RHS structural members for demountable industrial wall enclosure; allows controlled daylighting and future vertical extension.

Precast concrete vertical fins bolted to RHS steel members to reduce solar gain while maintaining ventilation.

Powder-coated CHS steel tube floor-to-ceiling grill providing passive ventilation, daylight penetration, and visual security.

Powder-coated steel railings bolted to steel deck floor for edge protection.

Exposed red-painted H-section steel members supporting elevated vending aisle bridge.

Powder-coated CHS steel railings bolted to concrete ramp floor for edge protection.

Steel casement folding doors enabling full opening for spatial fluidity and after-hours enclosure.

ELEVATION E1 SCALE- 1:200



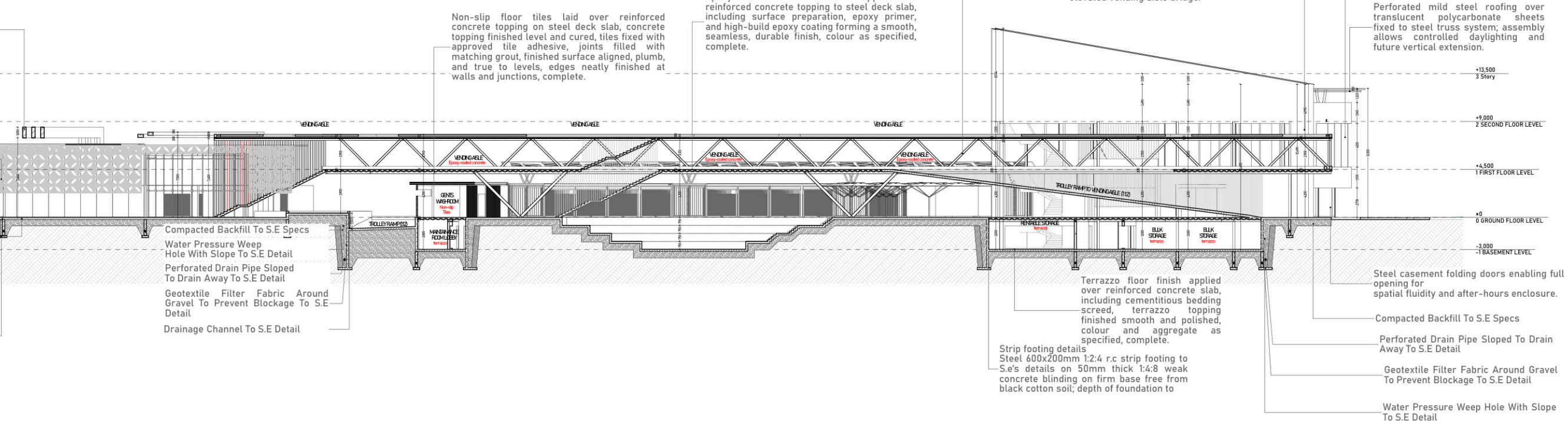
Water proofing aluminum Capping on Ndarugu stone masonry wall with light grey stucco finish defining primary entry node.

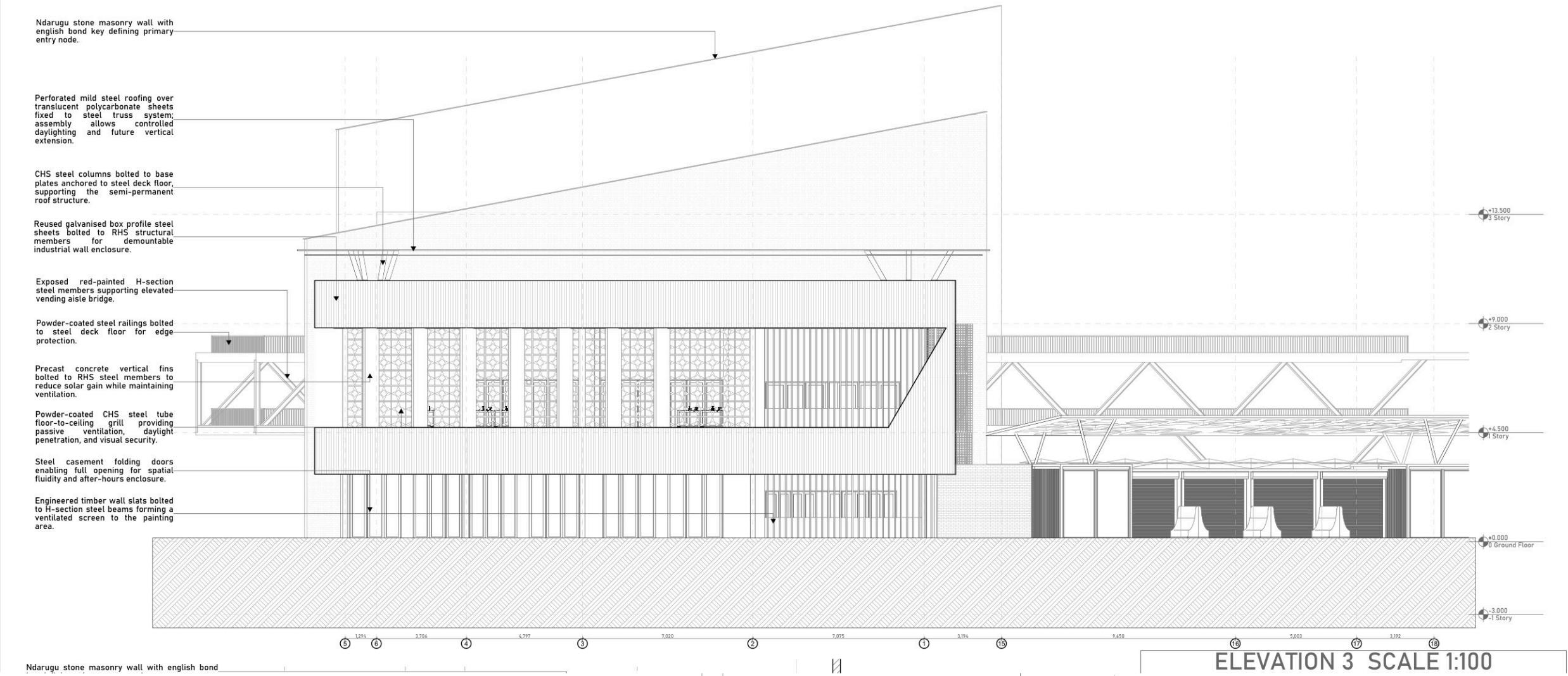
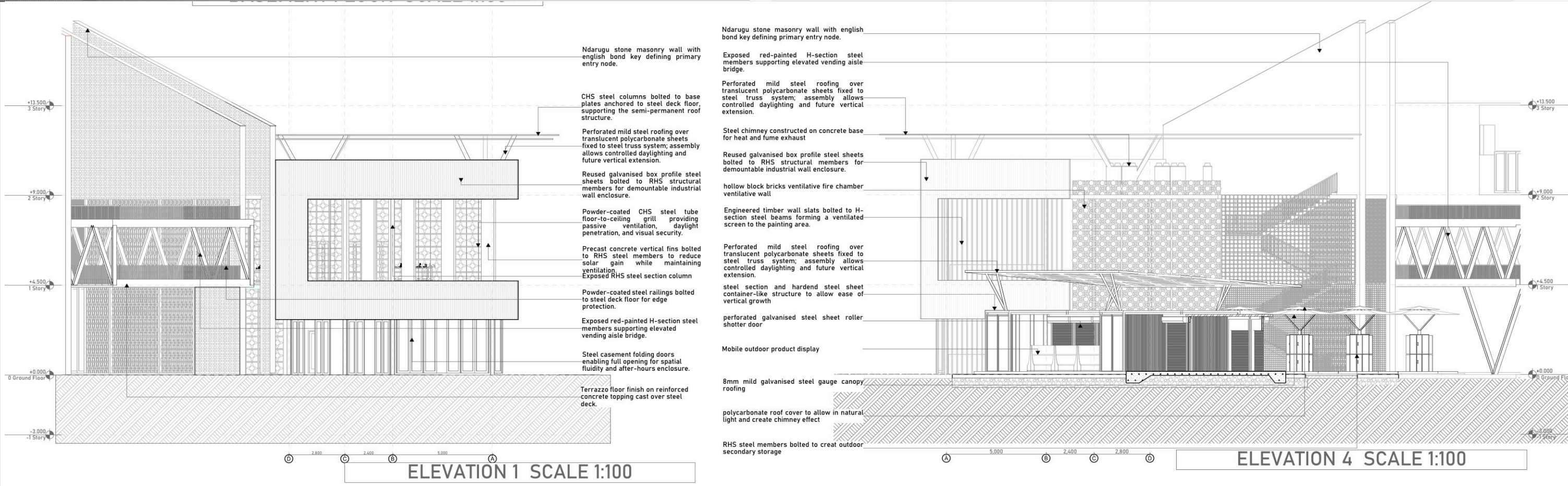
Perforated mild steel roofing over translucent polycarbonate sheets fixed to steel truss system; assembly allows controlled daylighting and future vertical extension.

Steel casement folding doors enabling full opening for spatial fluidity and after-hours enclosure.

Ground floor slab 150mm thick r.c slab mix 1:2:4 to structural engineer's details on 7.614258RC wire mesh on 1 layer of 1000g polythene sheeting, 300mm overlaps on 50mm sand blinding on 300mm well compacted approved hardcore on firm ground free from vegetable soil and treated with approved anti-termites insecticide.

Strip footing details Steel 600x200mm 1:2.4 r.c strip footing to S.e's details on 50mm thick 1:4:8 weak concrete blinding on firm base free from black cotton soil; depth of foundation to





Reinforced lightweight concrete blocks wall (100mm th)

Lightweight engineered soil mix (topsoil + compost + expanded clay / pumice)

Torch-on bituminous membrane (4 mm thick, root-resistant) applied over primed concrete slab and dwarf walls

R.C vertical sunshading element bolted to the rhs below

50 mm cement-sand screed or protection board over membrane

50 mm diameter PVC drain outlet connected to rainwater downpipe

Outlet fitted with leaf guard / gravel stop

secondary running 70 by 50 RHS section beam

50 by 75 rhs steel beam running along the wall cladding

50 by 75 rhs steel beam running along the wall cladding bolted to steel deck slab

150 mm thick steel deck slab, comprising profiled galvanised steel decking with reinforced concrete topping. Steel decking shall be 0.9-1.2 mm thick (18-20 gauge) or as structurally designed, fixed to supporting beams using approved mechanical fasteners or shear connectors to ensure composite action.

Reused galvanised box profile steel sheets bolted to RHS structural members for demountable industrial wall enclosure.

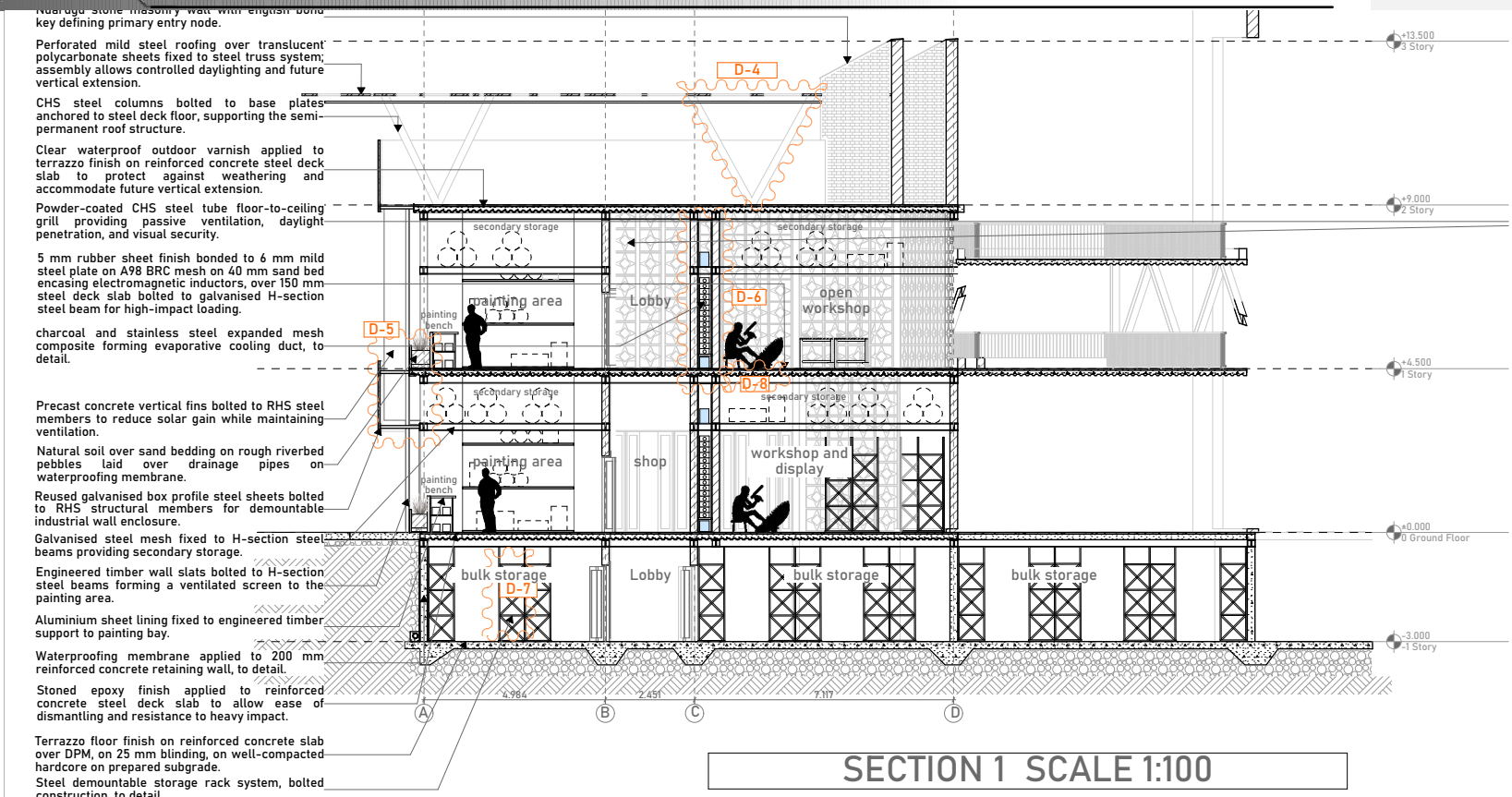
hot-dip galvanised H-section steel beams, designed to accommodate high-impact and dynamic loading. All connections shall be bolted or shear-connected in accordance with structural drawings and relevant standards, ensuring full load transfer and long-term durability.

Material pallet adoption: Use of mabati as a cladding to fit into context.

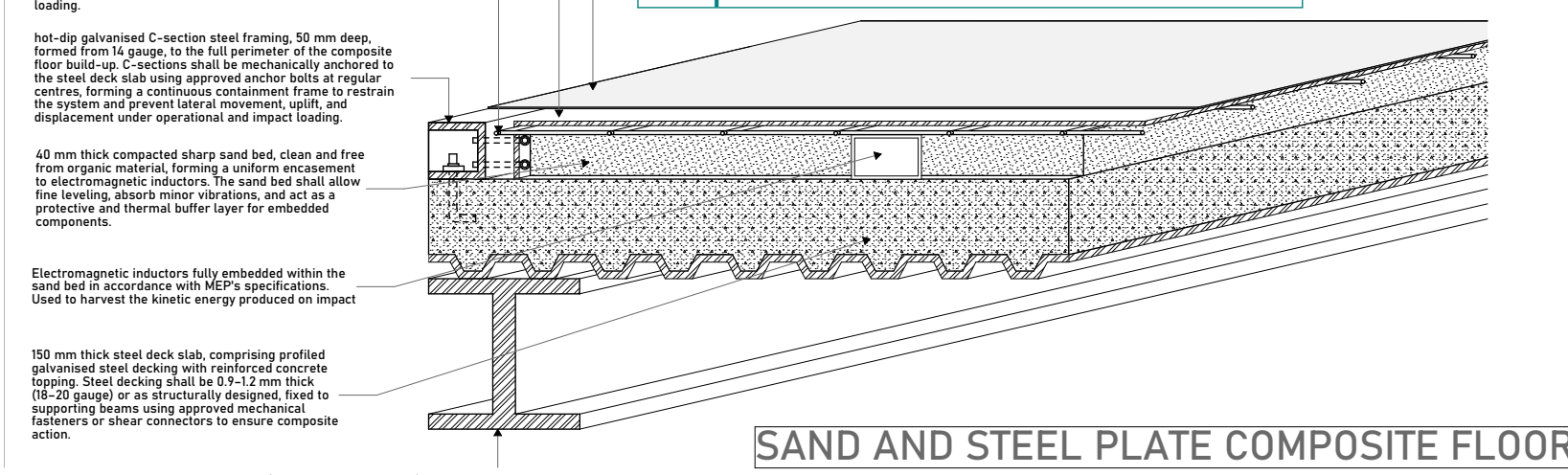
Light weight, demountable building systems: Use of materials like steel deck and steel section structres for easy replication, demounting or re-articulation.

KIGANDAINI INDUSTRIAL HUB

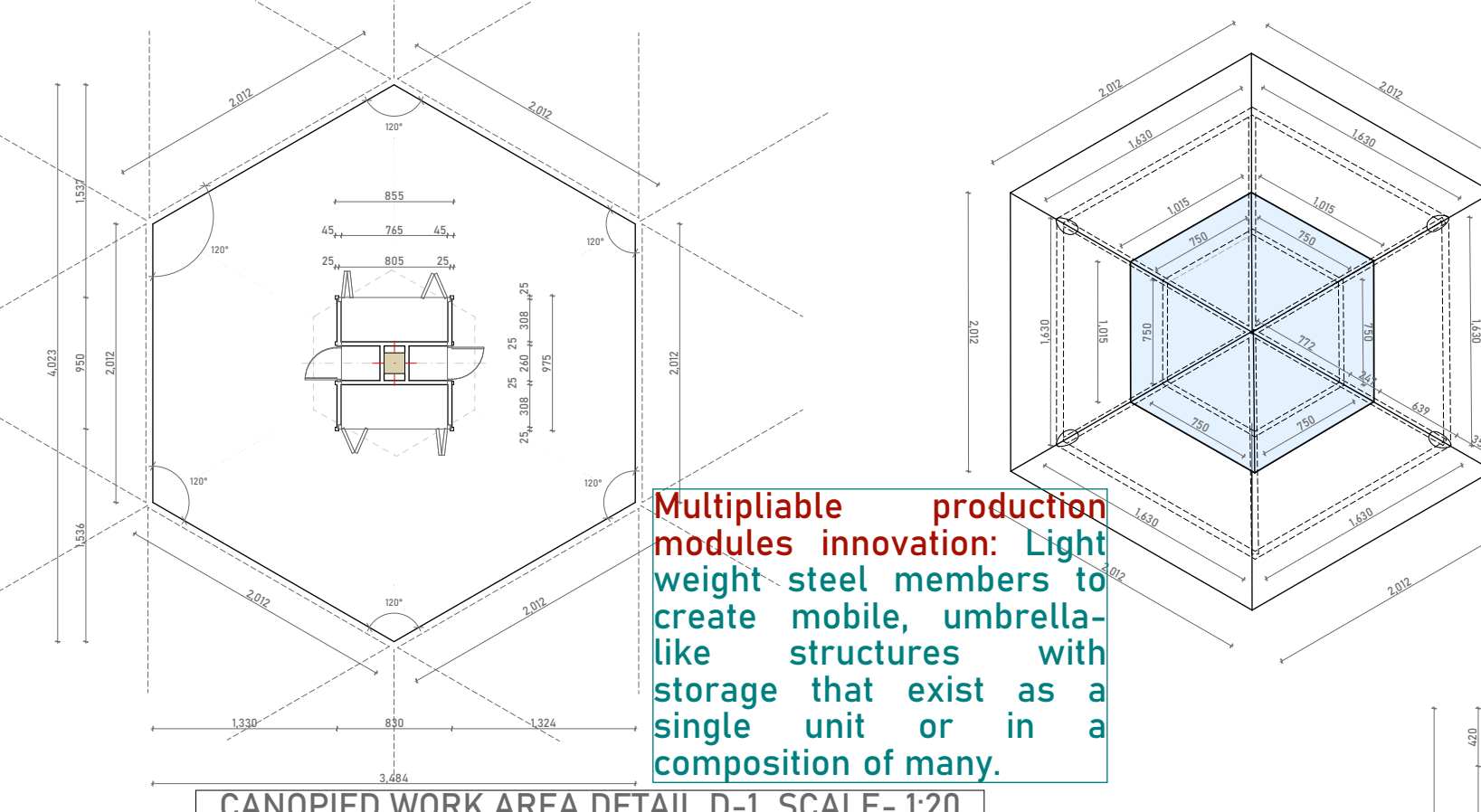
THE ROLE OF MATERIAL LOGIC ON SPATIAL OUTCOMES IN KENYAN JUAKALIS: A CASE OF THIKA KIGANDAINI



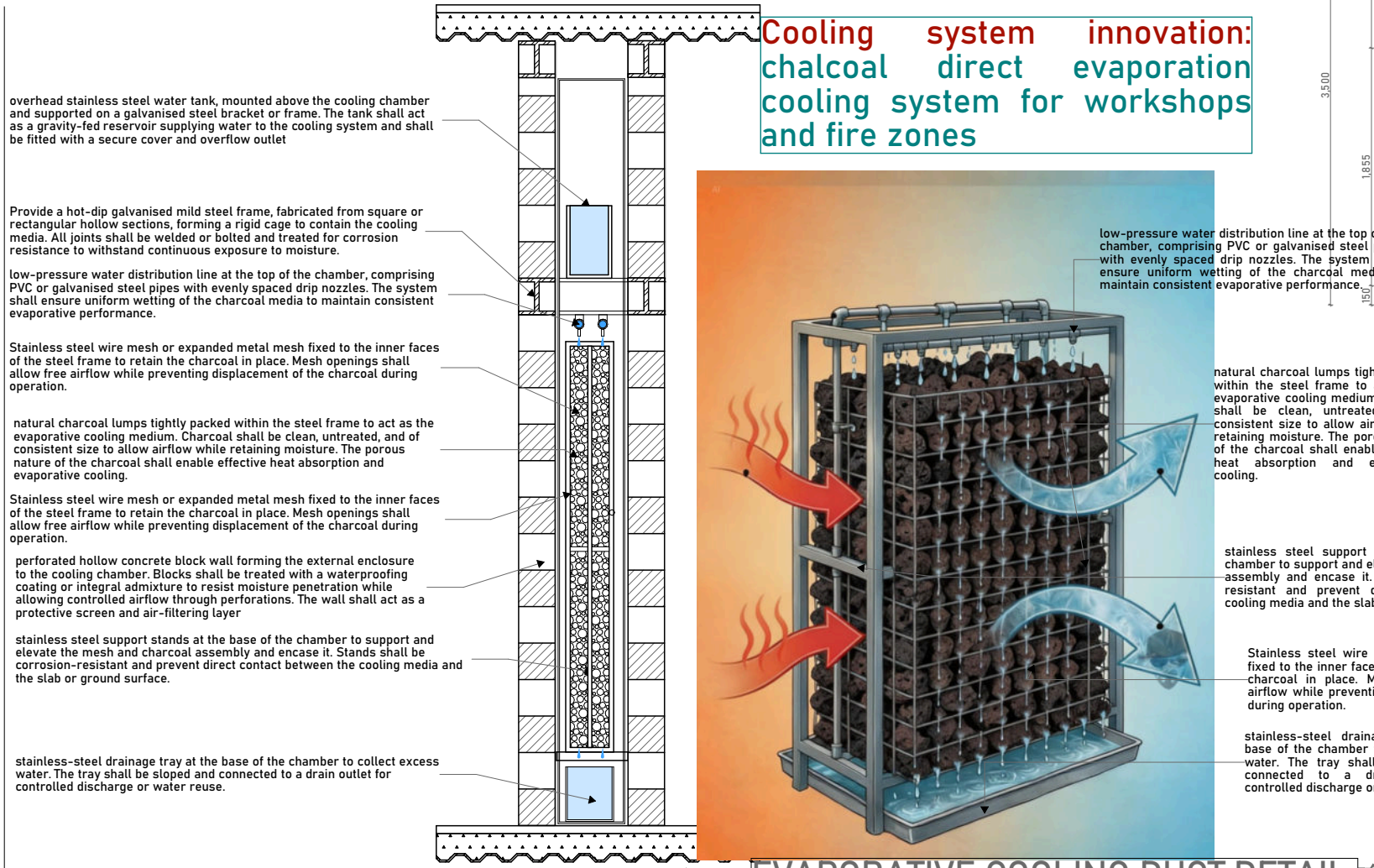
Material innovation: composite floor materials to absorb impact (sand, steel plate and mesh)
energy harvesting: electromagnetic inductors sandwiched between the composite material



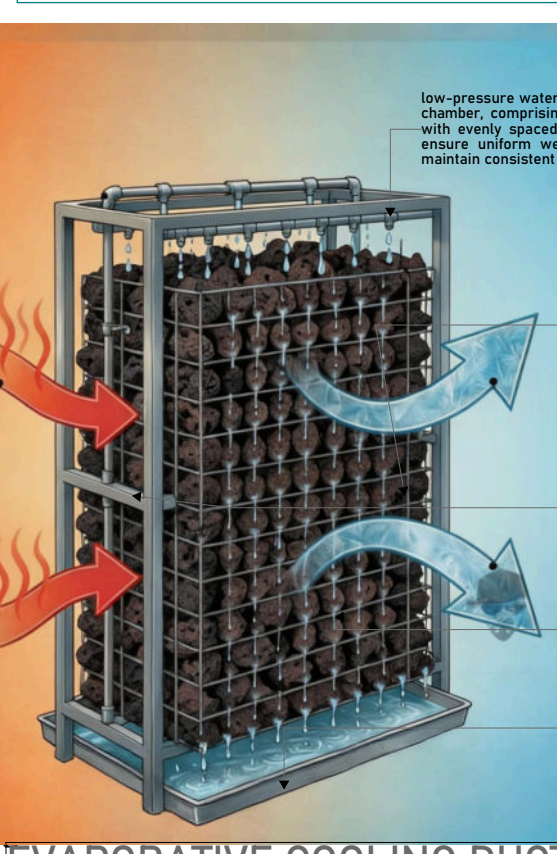
SAND AND STEEL PLATE COMPOSITE FLOOR



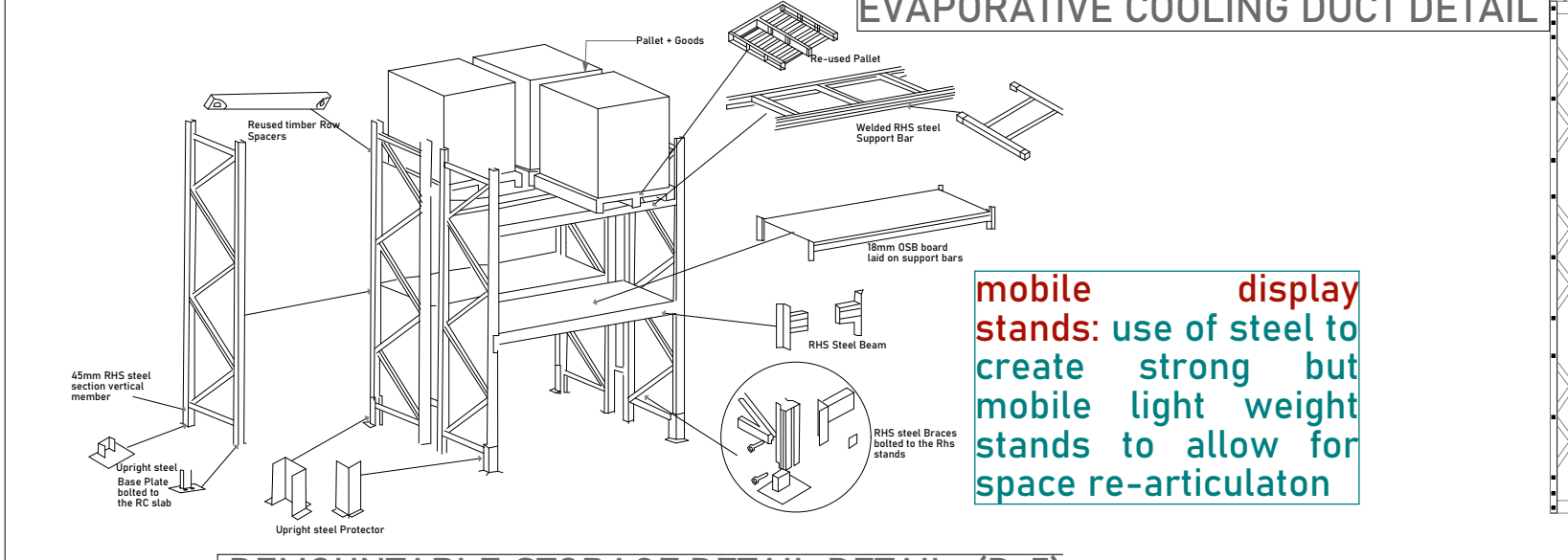
CANOPIED WORK AREA DETAIL D-1 SCALE- 1:20



Cooling system innovation: charcoal direct evaporation cooling system for workshops and fire zones

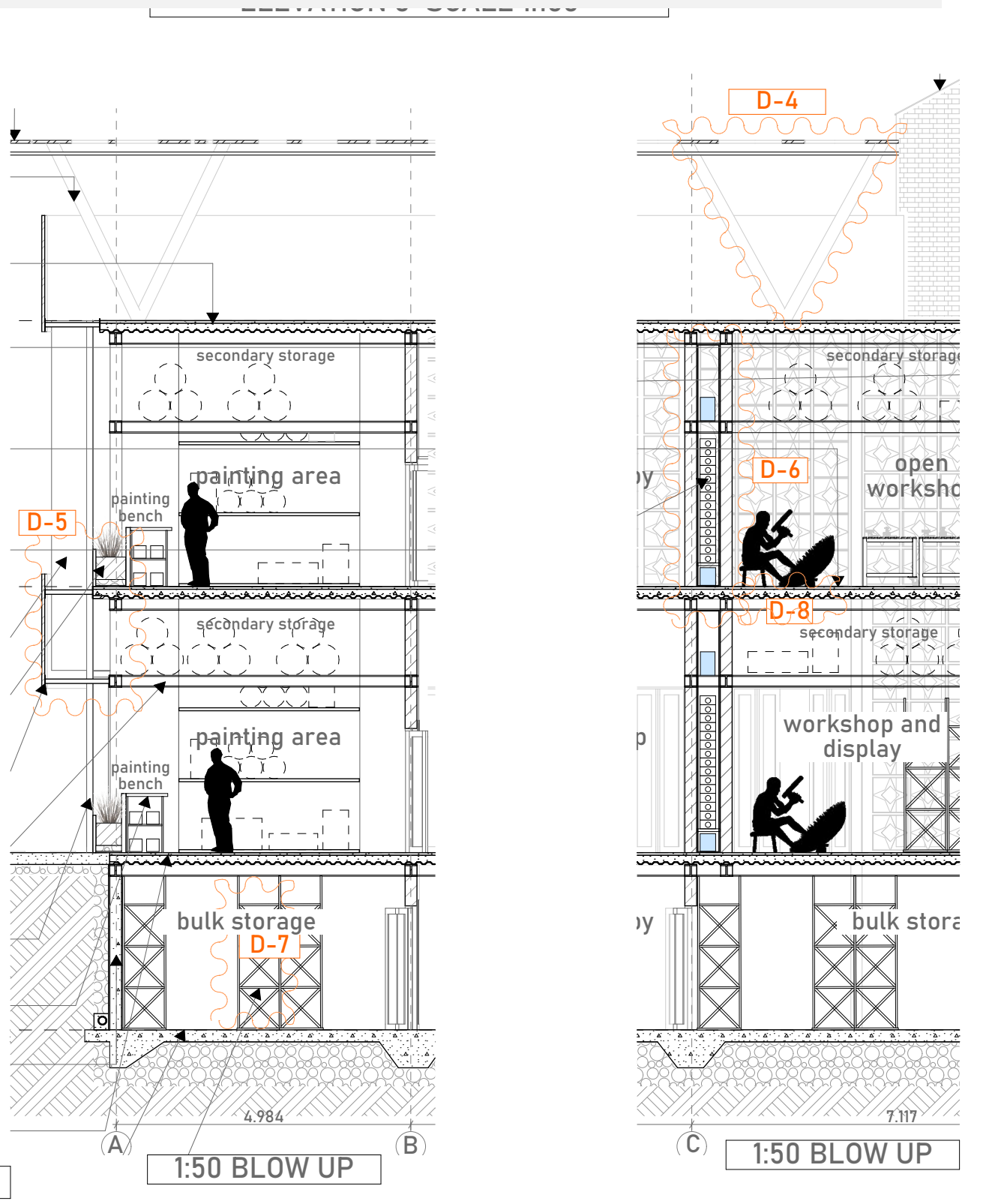


EVAPORATIVE COOLING DUCT DETAIL



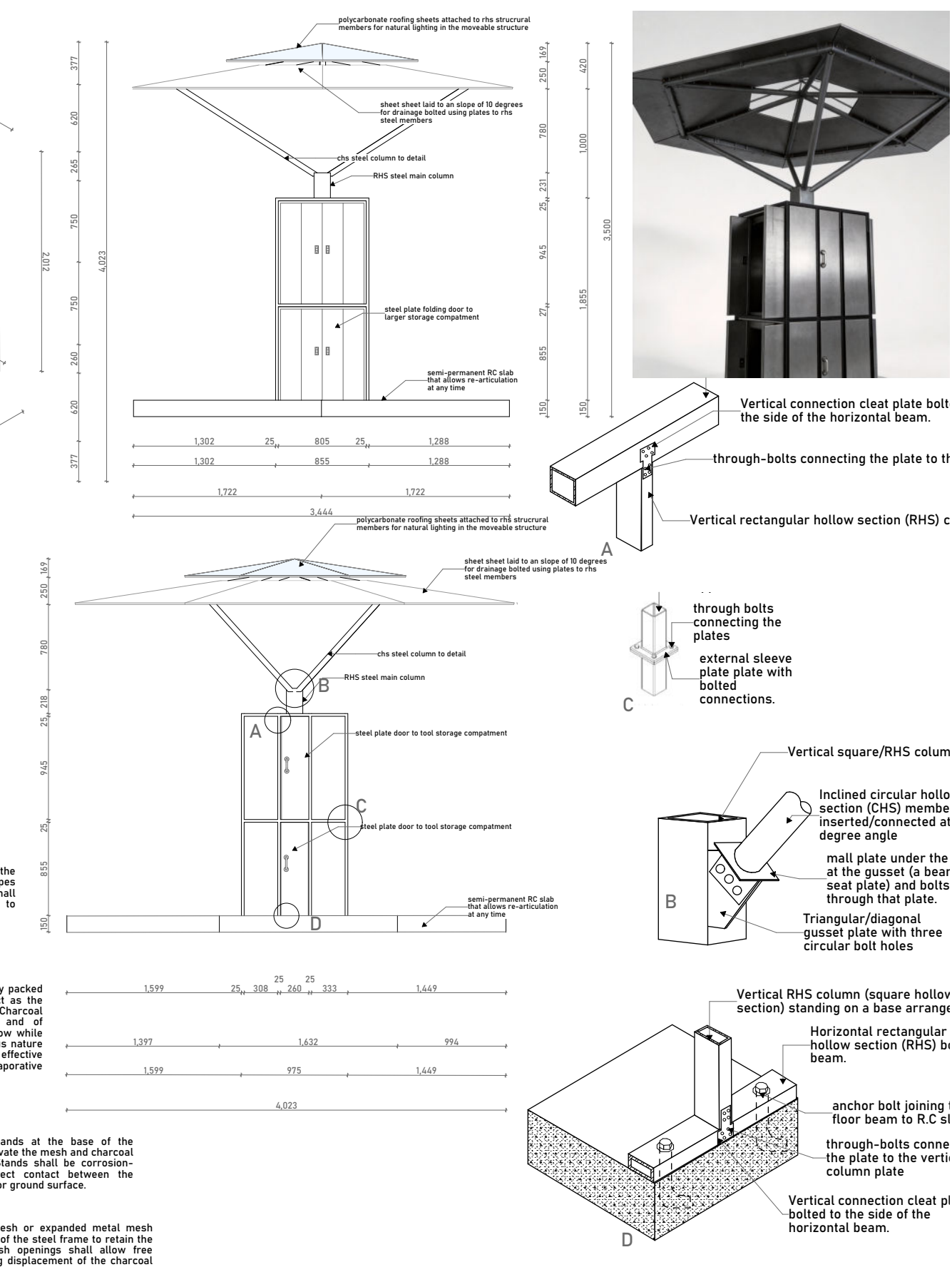
DEMOUNTABLE STORAGE DETAIL (D-7)

mobile display stands: use of steel to create strong but mobile light weight stands to allow for space re-articulation



1:50 BLOW UP

1:50 BLOW UP



RETRACTABLE/COLLAPSIBLE DOOR DETAIL (D-3)

territorial and deterritorial membranes: Use of steel retractable doors as membranes but also secure enough.

RESEARCH THESIS RECOMMENDATIONS IN DESIGN

EASE OF GROWTH

EFFICIENT LAYOUT SUPPORT

AESTHETICS AND FITTING INTO CONTEXT

LONGEVITY AND DURABILITY

COMFORT AND SAFETY



EXISTING + NEW
ARCHITECTURE
(RELATABILITY
THROUGH MATERIAL
USE)



ELEVATED AND
OPEN VENDING
AISLE



RHIZOMATIC REARTICULABLE AND
REPOPULATABLE STRUCTURES

COMFORT AND SAFETY

- USE OF MATERIALS THAT ALLOW NATURAL LIGHT PENETRATION WITH MINIMAL SOLAR GAINS
- USE OF FIRE SAFETY MATERIALS IN FIRE AREAS
- USE OF EVAPORATIVE COOLING-SUPPORTIVE MATERIALS.



**LONGEVITY AND
DURABILITY**

- USE OF HIGH LOAD BEARING MATERIALS ON WALLS
- USE OF ABBRESSION AND IMPACT RESISTANT MATERIALS ON THE FLOORS
- USE OF RIGID BUT EASILY REARTICULATED STUCTURAL MEMBERS



SOCIAL NODE AND TRAFFIC PULL AREA

EFFICIENT LAYOUT SUPPORT

- BLURRY LINE BETWEEN TERRITORIAL AND DETERRITORIAL ARCHITECTURE BY USE OF MATERIALS LIKE STEELTUBES, ENGINEERED TIMBER SLATES, STEEL RETRACTABLE AND FOLDING DOORS TO GAIN THE BENEFITS OF THE BOTH
- ENCOURAGING RHIZOMATIC OR ORGANIC GROWTH ON THE OUTDOORS THROUGH MODULAR STRUCTURE MATERIAL APPLICATION
- SEMI-PERMANENT PARTITIONS AND FIXTURES TO ALLOW FOR LAYOUT RE-CONFIGURATION



- ADOPTING AND IMPROVING EXISTING TACTICS ON
LAYOUT ARRANGEMENT



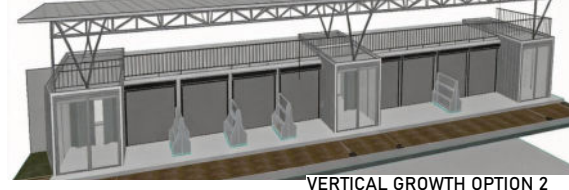
EXTENSION OF
MATERIAL LOGIC
TO SUPPORT
STRUCTURES

EASE OF GROWTH



VERTICAL GROWTH OPTION 1

- SEMI-PERMANENT ROOF STRUCTURES TO ALLOW FOR VERTICAL GROWTH
- USE OF SEMI-PERMANENT WALL PARTITIONS TO ALLOW FOR LATERAL GROWTH
- USE OF MATERIALS WITH EASY JOINERY TECHNIQUES TO ALLOW FOR EASY RE-ARTICULATION AND RE-POPULATION



VERTICAL GROWTH OPTION 2



- USE OF STEEL MODULAR RE-POPULATABLE UNITS DUE TO AVAILABILITY OF TECHNICAL KNOWLEDGE, ITS LIGHTWEIGHT AND EASE OF JOINERY AND DECONSTRUCTION PROPERTIES





THE END.
THANKYOU!

